Functional Versus Imperative Is Solved

Power, Cost, and Shared State in the Paradigm Debate

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1 Two Ways to Sum a List

Here is the same function written twice.

Listing 1: Imperative, in C.

```c
int total(const int *xs, int n) {
  int acc = 0;
  for (int i = 0; i < n; i++)
    acc += xs[i];
  return acc;
}
```

Listing 2: Functional, in Haskell.

```haskell
total :: [Int] -> Int
total []       = 0
total (x : xs) = x + total xs
```

The first is a list of instructions. The second is a pair of equations. Programmers have argued about which is better for most of the history of programming, and the argument is older than most of the languages it is now conducted in. We settle it.

Definition 1.1 (Imperative)**.** A program is *imperative* if its meaning is a sequence of commands, each of which changes a *store*, a map from locations to values. Running it executes the commands in order, and its result is read from the final store.

Definition 1.2 (Functional)**.** A program is *functional* if its meaning is an expression, and the meaning of an expression is its value. It is *pure* if the value of every expression depends only on the values of its subexpressions, so that any subexpression can be replaced by its value without changing the meaning of the whole. Strachey, borrowing from Quine, called this property *referential transparency* [14].

Remark 1.3. The term has since been used in several senses that are not equivalent [12]. We use Strachey's. Both definitions classify programs, not languages. C permits a function that assigns nothing, and Haskell permits one that does nothing but assign. Most languages in use permit both, so the question is not which language to use but which kind of program to write.

2 The Ceasefire of 1937

The first question, whether one style can compute something the other cannot, was settled before either had a machine to run on. Church proposed that the effectively calculable functions are those definable in the λ-calculus [2]. Turing proposed those computable by a machine that reads and writes symbols on a tape, following a table of instructions [15]. The first is a functional language and the second an imperative one.

Theorem 2.1 (The Church–Turing Ceasefire)**.** *A function on the natural numbers is λ-definable if and only if it is computable by a Turing machine.*

Proof. Turing showed that every λ-definable function is computable and every computable function is general recursive [16], and Kleene that every general recursive function is λ-definable [7]. □

Corollary 2.2 (No Exclusive Programs)**.** *Any function computed by a program in one Turing-complete language, functional or imperative, is computed by some program in any other.*

Proof. Each such language computes at least the functions that Turing machines compute, since it is Turing-complete, and at most these, since a Turing machine can run its interpreter. Theorem 2.1 is the case of the λ-calculus and Turing's machines. □

Remark 2.3. The ceasefire was signed in 1937, eleven years before the first stored-program computer ran a program [17]. It concerns which functions can be computed, not how quickly, how clearly, or how often incorrectly. The rest of this paper is about those.

3 Every Program Is Both

The ceasefire is not only a theorem. Compilers carry it out, in both directions, on most programs that are ever run.

Proposition 3.1 (Lambda, the Ultimate Goto)**.** *A call in tail position, whose result the caller returns unchanged, can be compiled as a jump that reuses the caller's stack frame. A tail-recursive function therefore runs as a loop, in constant stack space [13].*

A functional programmer who wants the loop of Listing 1 writes Listing 3, which GHC compiles to a loop.

Listing 3: Listing 1, as a functional programmer writes it.

```
total :: [Int] -> Int
total xs = go xs 0
  where
    go []       acc = acc
    go (y : ys) acc = go ys (acc + y)
```

Proposition 3.2 (SSA Is Functional Programming)**.** *In* static single assignment *(SSA) form every variable is assigned exactly once, and where control flow merges, a ϕ-function selects a value according to the block that control came from [3]. A program in SSA form can be read as a functional program: each block is a function, each ϕ-function a parameter, and each jump a tail call [6, 1].*

Listing 4: Listing 1 in SSA form.

```
entry:
  br loop
loop:
  i   = phi [0, entry], [i1, body]
  acc = phi [0, entry], [acc1, body]
  br i < n, body, exit
body:
  acc1 = acc + xs[i]
  i1   = i + 1
  br loop
exit:
  ret acc
```

Listing 5: Listing 4, read as functions.

```
entry()      = loop(0, 0)
loop(i, acc) = if i < n then body(i, acc)
               else exit(acc)
body(i, acc) = loop(i + 1, acc + xs[i])
exit(acc)    = acc
```

Clang compiles C to LLVM IR, which is in SSA form throughout [8], and the `mem2reg` pass promotes local variables whose addresses are never taken out of memory and into SSA form [9]. GCC has optimised in SSA form since version 4.0, in 2005 [4]. Arrays and other memory remain loads and stores, so Appel's reading covers the scalar variables [1], but the loop of Listing 1 spends most of its time in the compiler as the functions of Listing 5. C programmers have been writing functional programs for decades. The compiler was too polite to mention it.

The traffic runs the other way too. GHC translates Haskell through two functional intermediate languages, Core and STG, into Cmm, "a low-level imperative language with an explicit stack", and from there into machine code, directly or through LLVM [10]. Lazy evaluation, which in the words of Haskell's designers "kept us pure" [5], is itself implemented by mutation. An argument is passed unevaluated, as a *thunk*, and the first time its value is needed the thunk is overwritten in place with that value, so that it is computed at most once. Wadsworth proposed this in 1971 [18], and GHC's STG machine works this way [11].

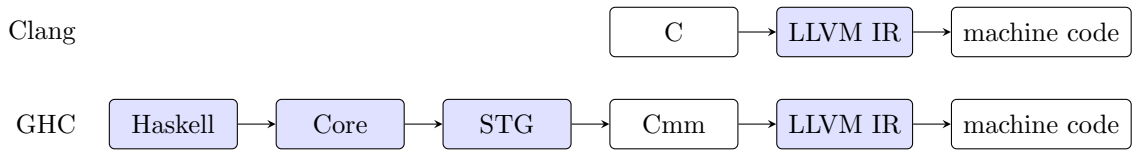


Figure 1: The stages of two compilers, with functional stages shaded: Clang, and GHC with its LLVM backend. Reading SSA form as functional (Proposition 3.2), the C program changes paradigm twice and the Haskell program three times.

Remark 3.3. The paradigm of a program is therefore a property of the stage at which a person reads it. Most of the debate concerns the first stage.

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